

# MADRE

Agentic  
System

• MODEL-AGNOSTIC • DELAYED • REASONING • EFFORT • [Beta v0.1]

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## Verification And Validation Plan

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# Quality Model

## 1.1 Quality Attribute Scenarios

| Attribute                     | Scenario                                                                                                                                       | Measure or evidence                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Responsiveness</b>         | The foreground remains conversational while deeper work is queued.                                                                             | Foreground latency and clear work status.                                            |
| <b>Recoverability</b>         | A failed or unsafe delayed task can be inspected and corrected.                                                                                | <code>RuntimeJournal</code> trace, status transition, and roll-back/correction path. |
| <b>Auditability</b>           | A result can be explained through context, <code>RuntimeEvent</code> evidence, action, <code>PolicyDecision</code> , and verification records. | Complete journaled outcome evidence chain.                                           |
| <b>Maintainability</b>        | An model backend or action contract can be replaced without changing kernel authority.                                                         | Contract and substitution review.                                                    |
| <b>Portability</b>            | A useful subset runs on ordinary local hardware.                                                                                               | Local profile and resource benchmark.                                                |
| <b>Security</b>               | Untrusted context cannot silently become instruction or policy.                                                                                | Prompt-injection and data-boundary scenarios.                                        |
| <b>Resource discipline</b>    | Delayed work is bounded by cost, time, storage, and compute policies.                                                                          | Queue and resource-use metrics.                                                      |
| <b>Privacy</b>                | Sensitive or unrelated local material is excluded from context and remote transfer unless scoped and policy-allowed.                           | Context exclusion and remote-transfer scenarios.                                     |
| <b>Model substitutability</b> | Boundary behavior is preserved across replaceable model-use profile and model backend choices.                                                 | Inference substitution scenario.                                                     |
| <b>Context isolation</b>      | One task's context does not silently bleed into another topic, user, inference execution, action invocation, or candidate.                     | Context scope and revocation review.                                                 |

| Attribute                                | Scenario                                                                                                                                                                  | Measure or evidence                     |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Explainability and inspectability</b> | Users and developers can see status, <code>PolicyDecision</code> , context evidence, <code>RuntimeEvent</code> evidence, action evidence, candidates, and recovery state. | Local read-first inspection scenario.   |
| <b>Testability</b>                       | Architecture claims can be checked through behavior scenarios and negative paths.                                                                                         | Requirements and decision trace review. |

# Evaluation And Benchmark Specification

**MADRE** defines an architectural evaluation surface for testing whether the runtime preserves local authority, governed context, bounded action use, evaluated learning promotion, journaled traceability, and recoverable delayed reasoning.

## 2.1 Benchmark Families

| Benchmark family                      | Purpose                                                                                                                                  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Functional correctness</b>         | Confirm expected runtime behavior for foreground-lane continuity and delayed work.                                                       |
| <b>Delayed reasoning quality</b>      | Evaluate evidence handling, decomposition, verification, and result delivery.                                                            |
| <b>Foreground triage</b>              | Confirm unsupported evidence-heavy answers are clarified or delayed rather than fabricated.                                              |
| <b>Context isolation</b>              | Confirm unrelated or sensitive knowledge does not enter the wrong bundle.                                                                |
| <b>Inference substitution</b>         | Confirm replaceable model-use profile and model backend choices preserve the same boundary behavior and expose failures.                 |
| <b>Policy enforcement</b>             | Confirm action, data, autonomy, authorization, and recovery rules are enforced.                                                          |
| <b>Bounded internal action</b>        | Confirm the bounded internal action is policy-gated, recorded, and harmless.                                                             |
| <b>Blocked unsafe actions</b>         | Confirm remote, host-command, filesystem, mutation, and critical-action attempts do not execute under bounded internal-action authority. |
| <b>Inference output non-authority</b> | Confirm generated material remains generated material, neither authority nor a candidate by itself.                                      |
| <b>Known-material truth control</b>   | Confirm a <code>KnowledgeRecord</code> with non-authoritative truth metadata is not used as factual truth.                               |

| Benchmark family                    | Purpose                                                                                                                                                     |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Learning promotion control</b>   | Confirm a <b>LearningCandidate</b> remains inactive until evaluated promotion and that saving <b>KnowledgeRecord</b> is not learning.                       |
| <b>Runtime journal traceability</b> | Confirm <b>ReasoningTask</b> , <b>RuntimeEvent</b> , <b>PolicyDecision</b> , and inference references expose lifecycle and recovery evidence locally.       |
| <b>Read-first observability</b>     | Confirm CLI/status and local console inspection expose lifecycle, policy, context, inference, action, knowledge/learning transition, and recovery evidence. |
| <b>Recovery</b>                     | Confirm failed or unsafe outputs can be inspected, corrected, deleted/detached, invalidated, superseded, or rolled back.                                    |
| <b>Resource use</b>                 | Measure latency, storage, queue time, model cost, and local compute use.                                                                                    |
| <b>Reproducibility</b>              | Preserve enough configuration, context, and event history to review outcomes.                                                                               |
| <b>Hallucination resistance</b>     | Confirm generated material does not become truth, knowledge, action authority, routing, behavior, or candidate status by itself.                            |

## Benchmark Catalogue

| Benchmark                    | Observed property                                                                                                             | Evidence                                                                |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Foreground response          | Conversational continuity under delayed work                                                                                  | Latency and interaction-state record.                                   |
| Delayed work durability      | <b>ReasoningTask</b> survives interruption and exposes status                                                                 | <b>RuntimeEvent</b> lifecycle in <b>RuntimeJournal</b> .                |
| Context isolation            | Only scoped material with suitable metadata enters a task context                                                             | Context bundle and <b>KnowledgeRecord</b> inspection.                   |
| Policy enforcement           | Disallowed action remains blocked or requires authorization                                                                   | Policy decision record.                                                 |
| Bounded internal action      | Bounded internal action remains local, policy-gated, and traceable                                                            | Action trace.                                                           |
| Inference output boundary    | generated material remains generated material, not authority or candidate by itself                                           | <b>RuntimeEvent</b> evidence, output-handling, and negative-path trace. |
| Knowledge truth authority    | Non-authoritative <b>KnowledgeRecord</b> is not used as factual truth                                                         | Truth-metadata use and correction trace.                                |
| Learning promotion control   | <b>LearningCandidate</b> remains inactive and storing <b>KnowledgeRecord</b> is not learning                                  | Promotion and negative-path trace.                                      |
| Runtime journal traceability | Local trace correlates work, events, policies, inference, and recovery                                                        | <b>RuntimeJournal</b> inspection trace.                                 |
| Read-first inspection        | Local console exposes status, trace, policy, context, inference, action, knowledge/learning transition, and recovery evidence | Inspection trace.                                                       |
| Recovery                     | Failed or unsafe result can be corrected or rolled back                                                                       | Recovery trace.                                                         |
| Model substitution           | model backend replacement preserves composed boundary behavior                                                                | Model-use boundary comparison.                                          |